

AI-assisted sensemaking: Human-AI collaboration for the analysis and interpretation of recorded facilitated conversations

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Figure 1: Small-group community conversation.

Abstract

In light of growing toxic polarization and societal fragmentation often fueled by social media, we are designing alternative communication spaces we refer to as *dialogue networks*—networks of people engaged in recorded small-group prompted dialogue. We introduce the dialogue network framework and our use of tools powered by large language models that assist humans in the analysis and interpretation of themes and patterns across conversations which we refer to as *sensemaking*. We pilot case studies in collaboration with community partners using a prototype AI-assisted sensemaking tool. Insights from these pilots can inform the use of AI for human-led community engagement processes.

CCS Concepts

• **Human-centered computing** → **User studies**.

Keywords

AI-assisted sensemaking, human-AI collaboration, facilitated dialogue, conversation analysis, large language models

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1 Introduction

Recent years have seen alarming levels of polarization witnessed on social media and even in public spaces in society where these spaces are becoming increasingly dominated by loud and extreme voices. Specifically, social media spaces are designed to optimize, in large part, for engagement in order to drive advertising revenue. This can and often does lead to unintended consequences, including the systematic amplification of emotionally provocative content

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without regard to accuracy or risk of harm [11, 12, 17], and makes social media a space that is not tailored for healthy and meaningful dialogue nor listening more broadly.

Motivated by this reality, we are designing alternative communication spaces which we refer to as *dialogue networks*—networks of people engaged in recorded small-group prompted dialogue that invites people to share their values, stories, and experiences in contrast to opinion statements that often divide us. The essence of dialogue networks is to provide a space based on trust where conversation participants feel empowered to speak and share personal stories and experiences which has been shown to elicit higher levels of understanding and empathy among participants (compared to the sharing of opinions) [9, 10]. The design of these networks is grounded in two main pillars: (1) ancient wisdoms such as dialogue, active listening, facilitation, trust and community building and (2) leveraging digital technologies (with a carefully designed consent architecture) which aid in the analysis, understanding and visualization of patterns in these conversations. Conversations are organized with various partners that have different goals and are facilitated by a trained facilitator to ensure the conversation covers certain topics or questions (based on a conversation guide co-designed with the partner organization) and to ensure that all participants get a fair chance at speaking time, thus at personal sharing.

We refer to the analysis—essentially making sense—of conversations as sensemaking which could involve a laborious human effort of highlighting salient parts of a conversation, generating a codebook that describes a set of conversations, applying the codebook (i.e., tagging the highlights with themes and topics) and synthesizing the patterns.

The rapid and impressive development of Large Language Models (LLMs) over the past two years has led to these models showcasing remarkable performance across many tasks that involve certain kinds of reasoning, content analysis and the generation and synthesis of text. To assist in the sensemaking process, we leverage LLMs to design a human-led AI-assisted sensemaking tool. We present in this work the different blocks in the sensemaking pipeline that this tool covers. Moreover, we present case studies in which we pilot this tool with community partners and describe insights from these pilots and lessons that can inform the use of AI in the design of community-interfacing digital systems.

2 Related Work

2.1 Facilitation and dialogue

Extensive research from the social sciences has emphasized the importance of dialogue for public understanding, especially as a way to address social conflict. This includes seminal work [3] that makes the case for the importance of prompted dialogue for collective understanding and highlights the distinct role the facilitator plays in establishing dialogue norms and steering the conversation towards effective sharing. Other work [14] emphasizes the significance of sustained dialogue for transforming human relationships amid entrenched conflict, while [13] highlights the role of storytelling in effective deliberative forums. Other works point out the impact of narrative for digital participation [4] and democracy [2].

2.2 Human-AI collaborative systems

Recent advances in large language models have led to an exponential interest in designing systems that both leverage these powerful tools and take human input. This highlights the power of combining human expertise and the analytical power of AI, which has been used to augment the evaluation of AI systems [15] as well as being used in many real-world applications such as healthcare [1, 7], gaming [6, 18], education [5, 8], accessibility [16] among others.

3 Research Team and Partners

3.1 CCC and Cortico

We are a team of researchers and engineers across two closely affiliated organizations: The **Center for Constructive Communication** (CCC) at the Massachusetts Institute of Technology and **Cortico** a social tech non-profit. CCC brings together a group of research staff, engineering staff, design staff, and graduate students from various backgrounds: human-computer interaction, computational social science, machine learning, natural language processing, digital interactive design, learning technologies, journalism, political science, cognitive science, mediation and conflict resolution. Informed by years of social media and media analytics, CCC’s work combines the ancient wisdoms of human conversation with emerging digital technologies to promote shared understanding and trust rather than reinforcing the “side-taking” and binary thinking that too often divides us. Cortico is a non-profit closely affiliated with CCC. Its vision is to become the platform that: surfaces unheard voices rather than the loudest; facilitates sharing experiences rather than battling opinions; and promotes empathy and connection rather than division and intolerance. Over the years, Cortico has cultivated a rich network of more than 200 partners across the US from cities and municipalities, schools, local community groups, and others.

3.2 Community partners

In this work, we present case studies showcasing pilots with two community partners: The **Maine Education 2050**¹ project (hereinafter referred to as *ME2050*) and the **Be Heard Durham** project² (henceforth referred to *Durham*).

ME2050 is an initiative led by Jennifer Chace, Assistant Director at the Center for Education Policy, Applied Research, and Evaluation at the University of Southern Maine. The project aims to engage various stakeholders (students, parents, educators, community members, and industry leaders) to imagine the Maine of the future. The focus of this initiative is schooling and exploring the role of schools in helping Mainers thrive personally, civically, and economically in their community. The ME2050 team that took part in our pilots had prior deep experience with qualitative coding and had moderate familiarity with AI. One of the unique challenges of this project is that multiple stakeholders were represented in the (conversation) data (students, parents, teachers).

¹<https://usm.maine.edu/cepare/maine-education-2050/>

²<https://www.durhamcommunityengagement.org/bhd>

The Durham project is an initiative by the City of Durham (North Carolina) that aims at collecting and amplifying voices of historically underrepresented communities in the City of Durham to help shape decision-making at the city level and lead to positive change. Durham members that engaged in our pilots had minimal familiarity with AI. The unique challenges of this project is that a large and diverse community was represented in the data and the discussions often involved very concrete and precise city planning issues.

4 AI-assisted Sensemaking: System Design

Once conversations are held, recorded and automatically transcribed, our goal is to assist human sensemakers in making sense of large amounts of conversation data. Beginning the codebook creation process is a daunting task for sensemakers. Often, hundreds of open codes require multiple iterative workshops and tagging iterations to become a finalized codebook. Our approach is motivated by providing AI-suggested codes to jumpstart codebook creation that humans can always steer, inspect, and evaluate.

In this section, we present the different stages that make up the sensemaking pipeline that was piloted in this work: (1) open coding, (2) codebook generation, (3) codebook application (tagging) and (4) generation of AI definitions.

An important note is that in these pilots, the input data we are working with is excerpts highlighted from the conversation transcripts—as opposed to complete transcripts. These highlighted excerpts (which we refer to as *highlights* are judged by human sensemakers as being salient and representing meaningful points in the conversation.

4.1 Open coding

In this stage, the goal is to generate open codes that describe the highlights. We are interested in capturing the rich nuances and often multiple themes covered in a single highlight and so we make sure to allow the LLM to capture this richness. To that end, we prompt a GPT-based LLM to generate 3 to 5 open codes that describe each highlight. The rationale behind this number of open codes is that we want to be able to cover the nuance mentioned above and so we do not want to restrict the model to generate a small number of open codes (i.e., less than 3) but, at the same time, when thinking of the next phase of consolidating these open codes into a coherent codebook, having an excessively large number of open codes might also affect the quality of the generated codebook. Iterations on the prompt involved specific word usage (e.g. open code vs topic vs theme) which seemed to give slightly different results in terms of the generated outputs. The final prompt is given in the appendix.

4.2 Codebook generation

In this stage, the goal is to consolidate all the open codes from the previous stage in order to generate a coherent codebook that is representative of the underlying conversation data and which can be used to tag the conversation highlights. Different considerations are at play here. We want a codebook to have two levels. The first level has high-level (parent) codes that represent the major themes that appear in a conversation collection. In the second level, for each of the parent codes, we find a list of sub-codes that represent more nuanced and fine-grained themes related to the parent code.

To generate a codebook, we experimented with two approaches: (1) a combination of classical clustering approaches combined with LLM prompting and (2) an approach based only on LLM prompting. In the first, we used k-means clustering to cluster the open codes and then gave the various clusters as input to the LLM and prompted it to synthesize those into a codebook. In the second approach, we experimented with various ways that are based only on LLM prompting. Our internal testing showed that the second approach yielded better results, leading to a codebook that is more representative of the conversation data. With the second prompting-only approach, we experimented in various ways to generate a structure of two levels as follows:

- (1) In a first request, prompt the model to generate high-level themes (parent codes). Then iterate over each of them, and ask the model to generate sub-themes (child codes) for each parent code.
- (2) In a first request, prompt the model to generate high-level themes (parent codes). In a second request, prompt the model to generate sub-themes for *all* parent codes (child codes).
- (3) In one request: prompt the model to generate a two-level codebook that fits the above description.

The reasoning behind these different approaches was the thinking that a divide-and-conquer approach might help the model in focusing on different parts of the codebook in explicit separate stages (as in the first two approaches above). However, our internal evaluation showed that despite the third approach being the most demanding (i.e., forcing the model to generate a whole codebook in one shot), it performed the best. This could be explained by the fact that, given the excellent ability of LLMs to synthesize data, the model was able to look at all the data, take into account the frequency and semantic similarity of different codes and produce a more coherent and more representative codebook than the first two approaches.

Adopting the third approach (generating the codebook in one call), we also experimented with different ways to vary the structure and form of the codebook including:

- (1) Generating a fixed-form codebook (fixed number of codes in parent level, fixed number of child codes).
- (2) Introducing language to minimize semantic similarity between codes when we noticed that sometimes some of the child codes across different parents were close in meaning.
- (3) Generating a free-form codebook (no upper limit on number of codes/subcodes) while including language around frequency w.r.t open codes (to generate a reasonably-sized and representative codebook)
- (4) Including partner/project goals and participant information in the prompt to personalize the codebook even more.

4.2.1 Internal evaluation. To assess the quality of the generated codebooks, four members of our research team evaluated AI-generated codebooks on two diverse community conversation datasets. Part of the evaluation focused on comparing the AI-generated codebook to one that was designed by a group of qualitative researchers. This was not supposed to mean that the AI tool should have reproduced the human-generated codebook in order for us to judge that the quality of the AI-generated codebook is acceptable. This was simply

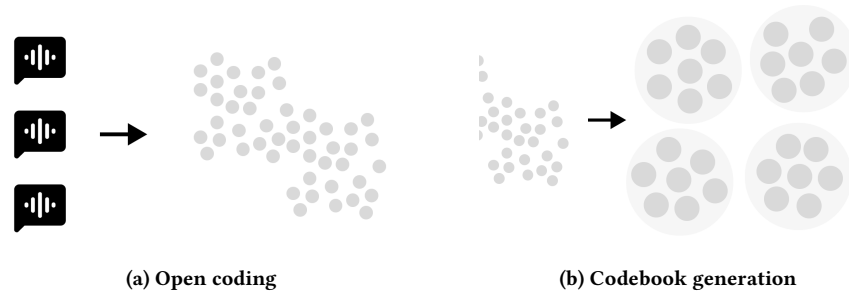


Figure 2: First two stages of the AI-assisted sensemaking process. (1) Open coding: Generating 3-5 themes/topics that describe a highlight. (2) Codebook generation: Consolidating all the themes/topics from the previous stage into a codebook.

carried out to compare and see how the two differ and whether the AI tool is generating a sensible codebook. The other part of the evaluation was data-driven. We mapped each of the open codes (generated in the previous stage) to one of the generated parent-code_child-code pairs in the codebook (essentially, via supervised classification). This allowed us to have a numerical view of how much each of the generated parent codes and child codes is supported by the open codes. In other words, this gave us a way to examine how much the AI-generated codebook was grounded in the conversation data and to what extent it represented and covered well the conversation data. This was also a way to compare different codebooks obtained via different LLM prompts.

4.3 Codebook application (tagging)

Given a codebook, thematic tagging involves identifying the proper codes that best describe a given highlight. As part of the proposed AI-assisted sensemaking tool, we provide AI-suggested codes (from the codebook) to the human sensemaker which they can then adopt or discard. We experimented with various ways in implementing the tagging stage. This includes iterating over each highlight and checking (classifying) whether each combination of parent/child codes applies or not. Another approach was to prompt the LLM by giving it the whole codebook and asking it, for each highlight, to return all combinations of parent/child codes that apply to the given highlight. Our internal evaluation showed that both approaches lead to a comparable accuracy performance. Given that the former approach is more resource intensive, we adopt the latter for the deployed system. The final prompt is given in the appendix.

4.3.1 Internal and external evaluation. To evaluate the AI tagging tool, four members of our research team manually tagged 200 highlights from a conversation dataset. This was used as a gold-standard to guide part of our design process when iterating over LLM prompts. Through this internal evaluation effort, we attempted to address the following questions:

- (1) Was each AI tag accurate?
- (2) Were all applicable tags applied?
- (3) How does AI tagging compare to human tagging?
- (4) What does AI do better? What do humans do better?

We also carried out an annotation effort with external evaluators through Prolific to further evaluate the tool on two different datasets.

4.4 Generation of AI definitions

As part of assisting human sensemakers in using better the codebook and applying the right tags, we also generate definitions for the themes and sub-themes given in the codebook. The generated definitions are not based only on the semantic meaning of the words that make up a given code, but they are also grounded in the conversation data. This makes sure that human sensemakers are given definitions that are both semantically and contextually appropriate. The prompt is given in the appendix.

5 Pilots with Community Partners

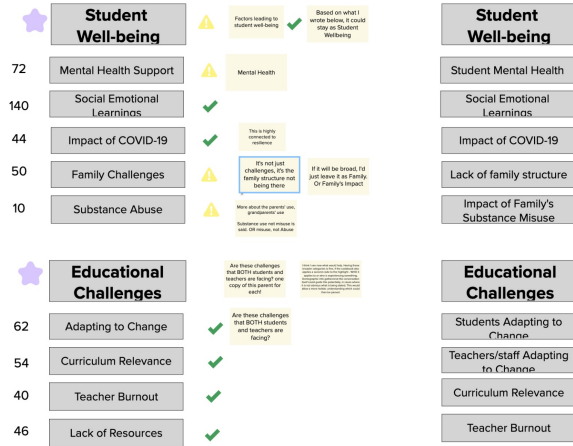
CCC and Cortico worked with two very different pilot partners. Some partners were deeply familiar with qualitative coding methods, others were familiar with how AI interplays with coding, and others were familiar with neither field of study. All brought both excitement and a mild dose of skepticism about involving AI in the sensemaking process.

5.1 AI-generated codebook pilot

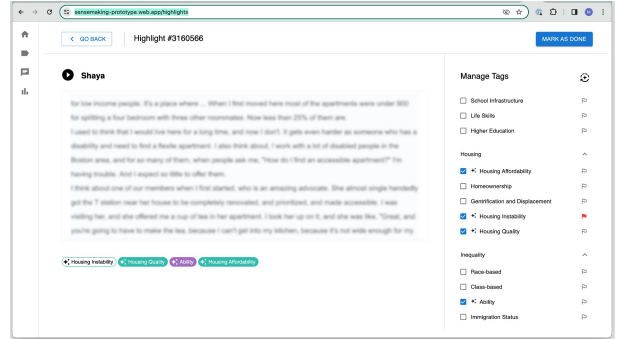
As mentioned in the previous section, one of the features of the codebook generation process is the possibility of including information about project goals and participants in the LLM prompt. Accordingly, in preparation for these pilots, we asked the partners the following:

If you had to prioritize three goals for this project, which ones would they be? Please focus on the top three goals for the project you have recently completed (not any future or aspirational goals for community engagement or community input in general). This specificity will help us tailor the AI-generated codebook so that we can review it next week with you.

Based on a set of conversations that the partner organization had already held, we generated a codebook with our tool. We designed a workshop to show members of the partner organization parts of the AI-generated codebook and ask for feedback. As a note, the partner team that participated in the workshop was intimately familiar with the conversation data and had highlighted themselves parts of the data. Figure 3a shows the mural where members of the partner team were asked to look at themes and subthemes and rate them and give feedback. Some of the research questions that our team had in mind when designing the pilot are the following:



(a) Interacting with AI-generated codebook.



(b) Tagging interface

Figure 3: Snapshots of various stages of the evaluation of the tool.

- (1) How do partners perceive and evaluate the **accuracy** of the AI-generated codebook?
- (2) What factors influence **user acceptance** and adoption across diverse organizational contexts?
- (3) To what extent can **bias be detected and mitigated** in AI-assisted sensemaking systems analyzing conversation data?
- (4) To what extent does this make the sensemaking process **more accessible to partners** with no/limited expertise in the analysis of conversation data?

Initial pilot results

We present here some of the initial results from this pilot. In Section 6, we present more detailed key lessons.

- (1) **Partners appreciated transparency** into the process of generating an AI-codebook, but they do not necessarily feel confident steering this process on their own just yet
- (2) **It is difficult for partners to effectively evaluate** an AI-generated codebook themselves, because humans cannot hold all key conversational data in mind.
- (3) Using an AI-generated codebook might **not necessarily reduce time** in the process of sensemaking, although it does shift the focus of the cognitive load and open up new opportunities (e.g., iterative qualitative coding while adding to the corpus of data).
- (4) **AI seems to reduce the impact of human biases**, although this is hard to tell in the moment of the workshop.

5.2 AI-assisted tagging pilot

As part of piloting our AI-assisted sensemaking tool, we designed an interface (shown in Figure 3b) through which the partner would see highlights along with AI-suggested tags (from the codebook). The user can approve/flag/reject tags, can see the reasoning for these tags (through the generated AI definitions) and can suggest additional tags themselves. Some of the motivating questions for this pilot were:

- (1) What do partners think about the tags suggested or missed by the AI tool?
- (2) If human sensemakers were to use the tool as a group, how would they view AI-suggested tags and assign them differently?
- (3) Is the reasoning (via AI definitions) helpful?
- (4) When shown numerical evaluations depicting how the AI-assisted tagging tool performed (in terms of accuracy) compared to the human evaluation (i.e. that of the partner), what are the partners' thoughts on tags that performed well or those that performed poorly?
- (5) Do partners feel the AI suggested tags add or detract from the sensemaking process?

6 Key lessons

We provide in this section key lessons that we arrived at following the pilots.

- (1) **What is the level of performance of the tool with a primary focus on accuracy and comprehensiveness?** Partners generally rated the codebooks positively; that said, their evaluations were offered usually with a grain of salt. Participants were more confident that they could evaluate more specific codes (e.g., "teamwork") than broader or more abstract codes (e.g., "empathy" or "inclusivity").
- (2) **How do partners perceive and evaluate the accuracy of the AI codebook?**

These partners instinctively compared the AI-generated codebook to a memory (or in some cases even their virtual copy) of their own human-generated codebook; similarities increased their faith in the AI-generated codebook, which was inevitable when they created their own version first.

When attempting to evaluate the accuracy of the AI codebook on its own merits, participants relied on their memories of the conversation and their best guess at the codes' definition. They recognized that this method was insufficient

for perfect evaluation, because no one could remember all the conversations (even if they were present for them and reviewed them recently, which was the best case scenario). Partners wanted example quotes to evaluate the language and organization of codes in the codebook.

Additionally, when participants knew that the codebook was AI-generated, anti-AI biases and feelings influenced their evaluations. This issue was more pronounced when the conversations centered around abstract ideas (e.g., empathy,) as compared to more concrete ideas (e.g., teamwork).

However, when participants were offered a numerical evaluation of every child code (i.e., number of times this AI-generated code was tagged by our tool), they were all eager to use this tool.

(3) How does the integration of AI impact sensemaking processes during the analysis of conversation data? What factors influence user acceptance and adoption across diverse organizational contexts?

All of our partners were relative AI-enthusiasts. A consistent driving force behind user acceptance was the wish to analyze more data while under systemic time constraints. If AI could start the work for them, our partners imagined that they would be able to sort through data much more quickly. They would also be able to do some thematic analysis while iteratively adding to the corpus of data, which two of our partners are continuing to do.

(4) What role does human-AI collaboration play in enhancing the overall sensemaking process?

When asked which parts of the sensemaking process should have AI involved, every partner agreed that they wanted AI involved in every step of the sensemaking except hosting the conversations with humans. At the same time, they wanted a well-informed human partner to review the AI's work and provide context. One issue that emerged for two partners was that the AI did not know who was the focus of the language in the conversations (e.g., teachers or students, employees or outside partners). This is the kind of high-level context that humans instinctively bring, but AI cannot deduce without clear labels, which are often missing in conversations.

(5) To what extent can bias be detected and mitigated in AI-assisted sensemaking systems analyzing conversation data? What strategies can be implemented to ensure fairness and impartiality in the outcomes?

The answer to this cannot be thoroughly answered within the scope of these pilots. However, we did notice one discrepancy in one of the pilots: human sensemakers sought to discard a child code ("need for rent control") under the parent code "housing affordability", whereas those with more qualitative analysis experience and the AI-generated codebook sought to keep it in the final codebook. Once the participants had a conversation about why AI might have included the code, they decided to keep it. In this way, AI helped mitigate human bias. Some partners were somewhat open to the idea of their perceptions being corrected by an AI-generated codebook.

(6) How can AI-assisted sensemaking tools reduce the cognitive load of analyzing conversation data?

Our partners are very eager to use the prototype to speed up the process of sensemaking and reduce the cognitive load as well as feelings of overwhelm. However, our current tools still require some support for most partners to feel confident in how to use them correctly or know what to do next.

Sometimes, parts of the AI-generated codebook do not feel right, but it is unclear how to steer it towards perfection. Some partners felt confident enough to make a few changes to the AI-generated codebook and then continue with this as their primary codebook for that conversation collection. Other partners spent a lot of time reviewing the AI-generated codebooks, but remained unsure of how to tweak that output to serve their needs going forward. From one point of view, their cognitive load was heavier in the end, given the addition of this second codebook to review.

7 Conclusion

We presented in this work the design and implementation of an AI-assistance prototype tool whose goal is to help human sensemakers analyze and understand community conversations at scale. We presented pilot studies with two partners who had varying goals and different composition of team membership.

We believe that AI has a tremendous role to play in this space although some challenges remain. Beyond the specific research questions above, we note that skepticism regarding AI assistance in the sensemaking process is highly linked to general feelings regarding AI as a technology and whether people should trust it in general. People are generally worried about the issue of privacy when using AI technologies and how issues of bias can affect the performance of such a tool and more generally the sensemaking process itself—which, for engaged members of community organizations, is supposed to be a personal and emotional process. Given the nature of community conversations that usually center around important issues for communities, partners had concerns and fears regarding misrepresenting a participant's voice. One could argue that a human process can still misrepresent a participant's voice, so this is also interlinked with the issue of AI skepticism. There's a strong feeling among partners that they want AI as thinking partner not the leader of the process, a partner that can help them identify patterns in the data that they may have otherwise missed and, in some cases, even identify some bias at the human side as we discussed in the paper.

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A LLM Prompts

We include in this section the prompts used for the various stages. We note that, in our work, we use OpenAI GPT-based LLMs. Over the span of the iterative process of implementing each stage and designing the prompts for each of the stages, OpenAI released upgrades of their LLMs. Accordingly, the process started with GPT3.5 (various upgrades) and then switched to GPT4 when it was released.

A.1 Open coding

You are provided with a quote from a project with [participants] aimed at [objective].

Your task is to summarize high-level ideas expressed in the quote with bullet points.

Instruction:

- Each bullet point should sound similar to the quote's text .
- Bullet points should explicitly capture what is being referenced and include the underlying purpose of why it was mentioned.
- Bullet points should not depend on the quote or other bullet points to be meaningfully understood.
- Do not repeat ideas between bullet points.

A.2 Codebook generation

1. Fixed-form codebook

The task is to generate a two-level codebook for a collection of conversations based on a list of [total number of codes] codes that represent themes and ideas that appear in these conversations. The codebook structure is two levels. The first level has 10 codes. These best summarize the high level ideas and themes that appear in these conversations. As a second level, each of the 10 codes of the first level has 5 sub-codes or sub-topics that are more fine-grained themes that are related to the parent code.

Codes:

[List of codes]

Based on the [total number of codes] provided codes, please generate a codebook with 10 parent codes with each of these codes having 5 sub-themes. Taken together, these 10 codes and their corresponding sub-codes represent the ideas and themes appearing in the conversations. Do not use more than 3 words per code.

Codebook:

2. Minimize semantic overlap

The task is to generate a two-level codebook for a collection of conversations based on a list of [total number of codes] codes that represent themes and ideas that appear in these conversations. The codebook structure is two levels. The first level has 10 codes. These best summarize the high level ideas and themes that appear in these conversations. As a second level, each of the 10 codes of the first level has 5 sub-codes or sub-topics that are more fine-grained themes that are related to the parent code. Avoid as much as possible having overlap or semantic similarity between parent codes and subcodes.

Codes:

[List of codes]

Based on the [total number of codes] provided codes, please generate a codebook with 10 parent codes with each of these codes having 5 sub-themes. Taken together, these 10 codes and their corresponding sub-codes represent the ideas and themes appearing in the conversations. Avoid as much as possible having overlap or semantic similarity between parent codes and subcodes. Do not use more than 3 words per code.

Codebook:

3. Free-form codebook

We are going to construct a codebook for a list containing hundreds of open codes that reflect what is being talked in a conversation collection. The codebook will have a two level structure: A high level where parent codes express the main themes and ideas talked about in the conversations and a lower level where each parent code has subcodes that are related to it and are more fine-grained and nuanced. I want a reasonable number of high level parent codes and a reasonable number of child codes. The number of parent codes and subcodes linked to each will depend on the distribution of the data so it is not expected that all parent codes have the same number of subcodes. One way to think about it is that a code should be represented in the codebook as a parent code if it appears at least 20 times in the list of open codes. The number of subcodes will vary depending on how prevalent a parent code and related themes appear in the list of open codes. Avoid as much as possible having overlap or semantic similarity between parent codes and subcodes.

List of open codes:

[List of codes]

Given the above list of open codes, please generate a two-level codebook. The number of parent codes and subcodes linked to each will depend on the distribution of the data so it is not expected that all parent codes have the same number of subcodes. Avoid as much as possible having overlap or semantic similarity between parent codes and subcodes. Do not use more than 3 words per code.

Codebook:

A.3 Codebook application (tagging)

You will be presented with a quote. Your job is to provide a set of tags for the quote.

Choose ONLY from the list of tags provided.

Possible tags (in 'Theme.Sub-theme' format):
[codebook]

Return the tags in a JSON array in this format:

```
[
  {
    "highlight_id": highlight_id,
    "tags": ["tag1", "tag2", ...]
  }
]
```

Only return the JSON array with no other notes or annotations.

A.4 Generation of AI definitions

List of open codes:

[List of codes]

As part of generating a codebook, we want to generate definitions for the each of the codes in this codebook. I want the definitions to be grounded in the list of open codes. I will first give you the codes and want you to generate a definition for each of them. The format is: parent code.child code

Codebook:

[codebook]

Now generate a definition for each of the codes given in the above codebook: